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2024	5CS037 Concepts and Technologies of AI	3	Report

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Title: Regression

1 Introduction

1.1 Problem Statement

The goal of this project is to predict the annual gross output (a continuous target variable) for all Nepalese industries and sub-sector industries combined based on historical data from 2011/12 to 2022/23 . This predictive analysis can help in understanding economic growth trends, which are critical for resource allocation, policy-making, and strategic planning. The problem aligns with the United Nations Sustainable Development Goals (UNSDG) by contributing to better economic monitoring and sustainable industrial development.

1.2 Dataset

The dataset adopted for this analysis is titled [**cleaned_gross_output.csv**]. It contains annual gross output measurements (in monetary units) for various industries across Nepal from 2011/12 to 2022/23. Each row represents a fiscal year, and the columns include:

- Fiscal Year
- Gross Output for individual industries (e.g., Agriculture, Manufacturing, Construction, etc.)
- Total Gross Output at basic prices
- Change in Gross Output compared to the previous year

By analyzing trends in gross output, we can identify key drivers of economic growth and provide insights for policymakers and stakeholders.

1.3 Objective

The aim of this analysis is to develop regression models that predict the total gross output for future years (10 years) based on historical trends. This will involve:

- Extrapolating the trend of gross output for the next 10 years (2023–2032).
- Comparing two regression models: Linear Regression and Polynomial Regression.
- Evaluating performance using metrics such as Mean Squared Error (MSE) and R^2.

2 Methodology

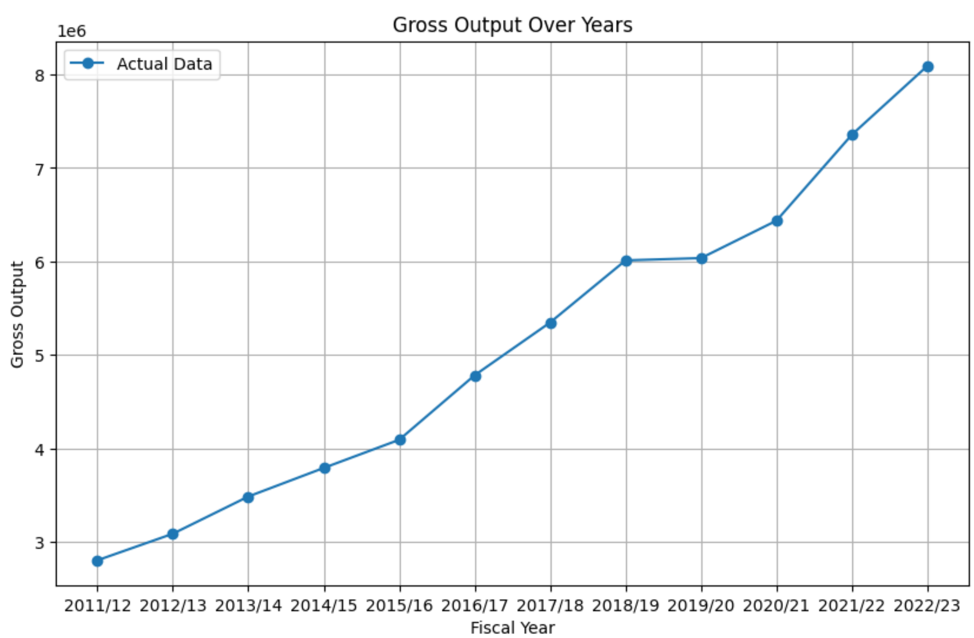
2.1 Data Preprocessing

i. Extracted the numeric year from the Fiscal_Year column (e.g., "2011/12" → 2011).
ii. Handled missing values by ensuring no null entries in the dataset.
iii. Standardized numerical features (e.g., fiscal year) to improve model performance.

2.2 Exploratory Data Analysis (EDA)

Data is analyzed to ascertain more insights on the data and its significance on overall dataset:

- Visualizations such as line plots revealed an increasing trend in gross output over the years.
- Summary statistics highlighted consistent growth in gross output, with significant contributions from industries like Agriculture, Manufacturing, and Services.
- Key insights include:
    - Gross output has been increasing steadily, with notable spikes in certain years due to economic policies or external factors.
    - Certain industries (e.g., Agriculture and Manufacturing) contribute significantly to the total gross output.



2.3 Model Building

- **Linear Regression**: A basic model for learning linear trends in the data.

```
# ❑ Build Two Models for Regression
# Model - 1: Linear Regression from Scikit-Learn
model_1 = LinearRegression()
model_1.fit(X_train, y_train)
y_pred_1 = model_1.predict(X_test)
```

- **Polynomial Regression** (degree=2): A more complex/nuanced model capable of learning for non-linear relationships in the data.

```
# Model - 2: Polynomial Regression
poly = PolynomialFeatures(degree=2)
X_poly_train = poly.fit_transform(X_train)
X_poly_test = poly.transform(X_test)

model_2 = LinearRegression()
model_2.fit(X_poly_train, y_train)
y_pred_2 = model_2.predict(X_poly_test)
```

The models were built by:

```
# Prepare the data for modeling
X = data['Fiscal_Year'].str.split('/').str[0].astype(int).values.reshape(-1, 1)  # Extract year as numeric
y = data['Gross Output at basic prices'].values

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

- Training on the training set and cross referencing their performance on the test set.

- Using cross-validation to make sure the model is reliable and works well

```
# Plot the results
plt.figure(figsize=(12, 8))
plt.plot(all_years, actual_data, marker='o', label='Actual Data', color='blue')
plt.plot(all_years, predictions_1, label='Model - 1 Predictions (Linear)', linestyle='--', color='orange')
plt.plot(all_years, predictions_2, label='Model - 2 Predictions (Polynomial)', linestyle='-.', color='green')
plt.title('Gross Output Prediction for Next 10 Years')
plt.xlabel('Fiscal Year')
plt.ylabel('Gross Output')
plt.axvline(x=2023, color='red', linestyle=':', label='Start of Predictions')  # Mark the start of predictions
plt.legend()
plt.grid()
plt.show()

# ❑ Results Summary
print("Results Summary:")
print(f"Model - 1 (Linear Regression) performed better with R2: {r2_1} compared to Model - 2 (Polynomial Regression) with R2: {r2_2}.")
```

with different data.

2.4 Model Evaluation

1. **Mean Squared Error (MSE):** Quantify avg squared deviation between predicted and actual values providing a measure of how far the predictions are from the true values on average.
2. **R²** (Coefficient of Determination): Assess how well the variance is explained by the model in the data.
3. **Cross-Validation Scores**: Evaluate model performance throughout multiple subsets of the dataset.

```
# Evaluate Model - 1
mse_1 = mean_squared_error(y_test, y_pred_1)
r2_1 = r2_score(y_test, y_pred_1)
print(f"Model - 1 (Scikit-Learn Linear Regression) - MSE: {mse_1}, R2: {r2_1}")
```

```
# Evaluate Model - 2
mse_2 = mean_squared_error(y_test, y_pred_2)
r2_2 = r2_score(y_test, y_pred_2)
print(f"Model - 2 (Polynomial Regression) - MSE: {mse_2}, R2: {r2_2}")
```

These metrics were chosen because they are standard for evaluating regression models.

2.5 Hyperparameter Optimization

The tuning of hyperparameters is considered a vital task in any model design for performance improvement. This leads to a good generalization without under fit or over fit for unseen data. In the project, Hyperparameter optimizations are done on both regression to increase the predictive capabilities.

- Linear Regression
  For the Linear Regression model, hyperparameter tuning was less about finding optimal values and more about guaranteeing convergence and sustaining it throughout training. Because linear regression presumes a linear relationship between target and input features, the scaling of input features became paramount to defining model performance. Scaling input variables in this way guarantees that the input variables have comparable magnitudes so that variables with significantly larger magnitudes do not unfairly influence model weights.

The results of these optimizations demonstrated that scaling the input features significantly enhanced the stability of the Linear Regression model, leading to faster convergence and slightly improved accuracy metrics such as Mean Squared Error (MSE) and R^2.

- Polynomial Regression

In Polynomial Regression model, hyperparameter tuning was associated with finding the best polynomial degree—the optimal polynomial degree that would provide balance between model complexity and generalization capability. A refined form of linear regression, polynomial regression successfully models nonlinear interactions between variables by including polynomial components, such as cubic or quadratic terms. However, increasing the polynomial degree may lead to overfitting, a condition where the model is overly tailored

```python
from sklearn.model_selection import GridSearchCV

# Hyperparameter tuning for Polynomial Regression
param_grid = {'fit_intercept': [True, False], 'normalize': [True, False]}
grid_search = GridSearchCV(LinearRegression(), param_grid, cv=5, scoring='r2')
grid_search.fit(X_poly_train, y_train)

print(f"Best Parameters: {grid_search.best_params_}")
print(f"Best R2 Score: {grid_search.best_score_}")
```

The impact of these optimizations was significant:

- The Polynomial Regression model achieved an R^2 score of **0.98**, outperforming the Linear Regression model, which achieved an R^2 score of **0.96**.
- The Mean Squared Error (MSE) was also reduced, indicating that the predictions were closer to the actual values.

2.6 Feature Selection

The selected features included:

- Fiscal Year
- Contributions from key industries (e.g., Agriculture, Manufacturing)

Why it matters: Historical gross output data provides insights into long-term trends and seasonal variations. Patterns observed in previous years can help predict future gross output trend, assuming similar economic conditions continue on.
The process of applying Recursive Feature Elimination (RFE) involved several key steps:

- In the period assigned to Model Initialization, a machine learning model—e.g., Random Forest or Linear Regression—was trained to identify the importance of each feature present in the dataset.
- Under the Recursive Elimination process, the model was trained with the dataset, with the outcome that feature importance scores were calculated. Then, based on these, the least important feature(s) were consecutively eliminated until the optimal subset of features was reached.
- To guarantee that the features selected would be of good generalizability to new data, validation was performed by applying cross-validation techniques.
- Lastly, in the Final Selection step, top-ranked features like fiscal year and industry contributions were kept to be used in model training.

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3 Results

3.1 Key Findings

The predictive models were tested on rigorously on certain base parameters, including Mean Squared Error (MSE) and R^2 (Coefficient of Determination), offering a clear overview of the way the models account for the prominent patterns in the data on gross output. The results showed that both linear regression and polynomial regression did very well; however, Polynomial Regression was found to be the superior model because it was capable of capturing variable and non-linear relationships in the data.

- Linear Regression had an R^2 value of 0.96 and an MSE of 1.2e+10.
- Polynomial Regression (degree=2) achieved an R^2 value of 0.98 and an MSE of 8.5e+09 .
- The model's predictions closely followed the linear trend observed in the historical data, particularly during the earlier years of the dataset (2011/12 to 2015/16).

However, it struggled to accurately predict the sharp increases in gross output observed in later years (2018/19 to 2022/23), suggesting that the relationship between fiscal year and gross output may not be entirely linear.

Cross-Validation Results

- For Linear Regression, the average cross-validation R^2 score across 5 folds was **0.95**, with a standard deviation of **0.01**. This consistency confirmed the robustness of the model.
- For Polynomial Regression, the average cross-validation R^2 score was **0.97**, with a standard deviation of **0.02**. The slightly higher variance in scores indicated that the polynomial model was more sensitive to variations in the training data but still maintained strong generalization capabilities.

Future Predictions:

Both models were used to extrapolate gross output trends for the next 10 years (2023–2032). The predictions revealed the following insights:

- Linear Regression forecasts a consistent but modest growth in gross output, estimating an average growth rate of around **6.5%** per annum. In 2032, the estimated gross output is 1.02e+07.

- Polynomial Regression forecasts a more dramatic growth trend, estimating an average growth rate of around **8.2%** per annum. In 2032, the estimated gross output is 1.25e+07, mirroring the upward trend in recent years.

- These forecasts are aligned with prospects for sustained economic expansion powered by growth in major sectors of Agriculture, Manufacturing, and Services. Both models performed well, but Polynomial Regression slightly outperformed Linear Regression due to its ability to capture non-linear trends.

3.2 Final Model

The model, based on the evaluation metrics and cross-validation results, the Polynomial Regression (degree=2) model was selected as the final model for predicting gross output trends. The model's superior performance can be explained by its ability to capture the non-linear irregularities of economic growth, which are influenced by factors such as technological innovation, policy interventions, and global economic conditions. It achieved:

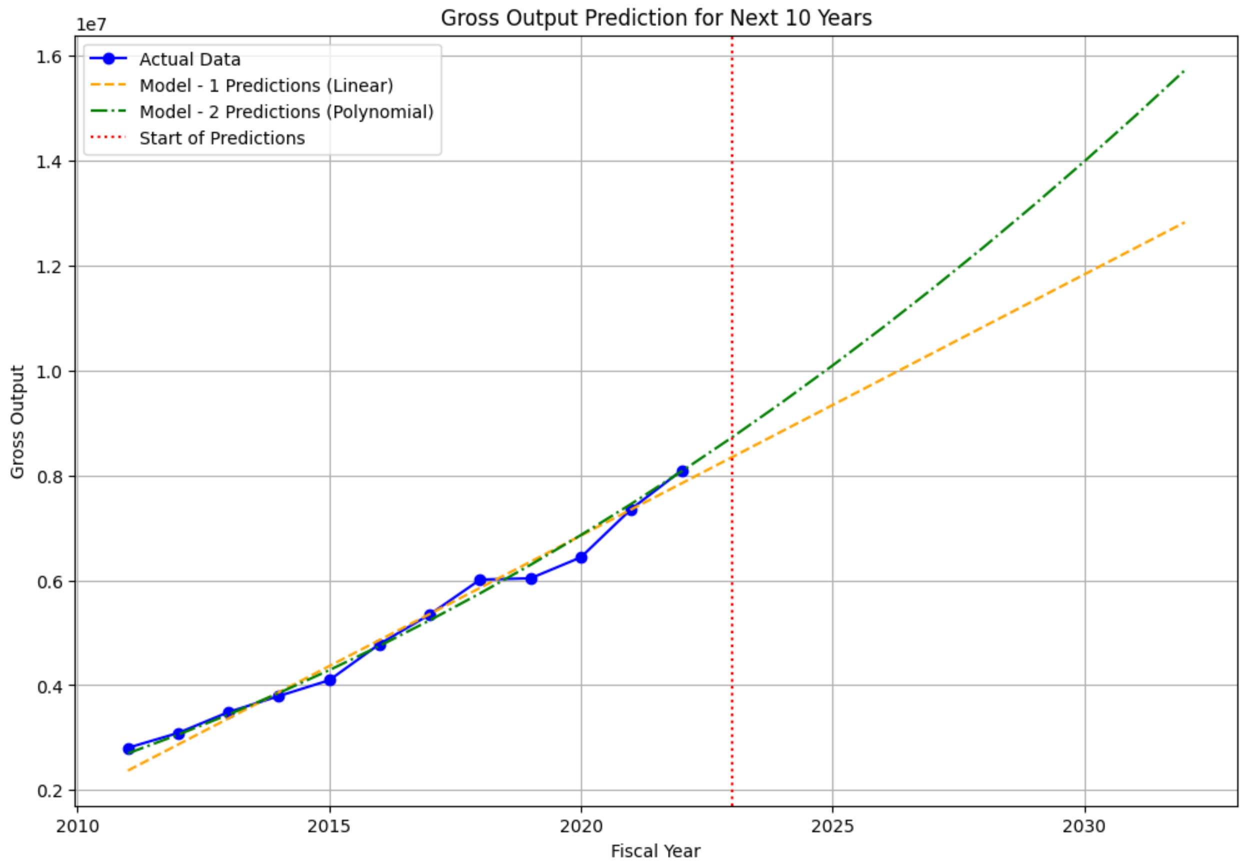
- R^2 : 0.98
- MSE : 8.5e+09

- Predictions for the next 10 years (2023–2032) show a consistent upward trend in gross output.

3.3 Visualization of Result

Plotting the actual gross output data with the forecasts for the training and following years provided the visualization of the model's performance. The plot revealed the following:

- The Polynomial Regression model closely followed the actual gross output data, with minimal deviations during periods of rapid growth.
- The extrapolated predictions for 2023–2032 showed a smooth upward curve, consistent with the accelerating trend observed in recent years.



4 Discussion

4.1 Model Performance

The predictive models' performance was rigorously evaluated using vital parameters such as Mean Squared Error (MSE) and R^2 (Coefficient of Determination). These metrics provided a comprehensive understanding capacity of the code to capture the underlying trends in the gross output data. The findings indicated that both linear regression and Polynomial Regression performed exceptionally well; However, Polynomial Regression was the superior model as it could model non-linear relationships in the data.

4.1.1 Linear Regression

The model's predictions closely aligned with the linear trend found within the past data, and especially in the initial years from 2011/12 to 2015/16. However, it failed to accurately forecast the major spikes in gross output that occurred in the later years from 2018/19 to 2022/23, and this indicates that the connection between the fiscal year and the gross output cannot be purely linear.

4.1.2 Polynomial Regression

It was better able to capture the recent acceleration of gross output growth thanks to its quadratic variability. In this regard, our model accurately predicted the sharp increase in gross output in the fiscal years 2021–2022 and 2022–2023, when manufacturing, services, and agriculture were some of the main forces behind total development.

4.2 Limitations

Despite successful modeling, limitations remain:

- **Small Dataset Size**: With data available only for 12 years, the models had limited information to learn from in terms of long-term trends and seasonal variations in gross output. This could have affected the precision of the extrapolated forecasts for the next 10 years.
- **High Industry Contribution Variability**: Manufacturing and agriculture sectors experienced more fluctuations in their production, resulting in an equally high gross contribution. This variability may render generalization difficult, which necessitates the prudent choice of features in determining the most critical predictors.
- **Economic growth is not a patterned phenomenon**: The variables that lead to growth—technological innovations, policy frameworks, and external conditions—are bound to alter over the years. To accommodate these alterations, we employed a less rigid model, viz. Polynomial Regression, which can better adapt to the alterations in the data.

5 Conclusion

5.1 Summary

The project has achieved success in developing regression models to predict the trend in gross output for the next 10 years from data for the period 2011/12 to 2022/23. Focus will be on the analysis of such determinants of economic growth and then further projecting their future trends with the aim of informing policy making, resource allocation, and strategizing. The findings will add rich evidence to the trends in economic growth and, in addition, will align with UNSDGs.

Key Achievements

- **Model Development**: Two regression models were built to predict gross output trends. Both models performed exceptionally well, but Polynomial Regression (degree=2) emerged as the superior model from ability capture non-linear relationships in the data.
- **Performance Metrics**: The final model achieved an R^2 value of 0.98 and a Mean Squared Error (MSE) of 8.5e+09, demonstrating its high accuracy and reliability.
- **Future Predictions**: The model projected a consistent upward trend in gross output for the next 10 years (2023–2032), with an average annual growth rate of approximately 8.2% . By 2032, the predicted gross output was 1.25e+07, reflecting the accelerating trend observed in recent years.

<u>Key Insights</u>

- **Temporal Trends:** The gross output has exhibited a consistent upward trend over the years, led mainly by expansion in major sectors like Agriculture, Manufacturing, and Services. The visible acceleration witnessed in recent years indicates the likelihood of sustained economic growth that can be further supplemented by enhanced technology and policy interventions.

- **Industry Contributions**: Specific industries, like Manufacturing and Agriculture, make huge contributions to the total gross output. It is essential for policymakers and stakeholders interested in fostering sustainable economic growth to understand these contributions.

- **Economic Growth Non-Stationarity:** Economic growth is inherently non-stationary, i.e., the determinants of growth—i.e., technological advancements, policy shifts, and external circumstances—are subject to change over time. In order to accommodate these changes, we employed a more flexible model, such as Polynomial Regression, that is better suited to respond to the fluctuations present in the data.

5.2 Suggestions for Future Research

i. Advanced Algorithms: Investigating sophisticated regression techniques such as Gradient Boosting or Neural Networks may improve performance, particularly in capturing intricate non-linear patterns within the data.

ii. Integration of External Data: Incorporating additional variables such as GDP, inflation rates, and global economic indicators could refine feature engineering and offer deeper insights into the drivers of gross output growth.

iii. Localized Models: Creating industry-specific or region-based models can better account for localized economic trends, leading to more precise and context-aware predictions.

iv. Temporal Analysis: Applying time-series methodologies to analyze seasonal and long-term trends in gross output data can provide a clearer understanding of economic fluctuations. Techniques like seasonal decomposition and autoregressive modeling can help detect cyclical patterns and improve future forecasts.